SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I P.Vishwa Nandan Reddy(192473015) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: vishwanandan

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition: Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet, usually on a pay-as-you-use basis.

Four characteristics of cloud computing:

1. **On-demand self-service:** Users can access computing resources whenever needed without contacting the service provider.
2. **Broad network access:** Cloud services can be accessed through the internet using computers, laptops, tablets, and smartphones.
3. **Resource pooling:** The provider shares computing resources among multiple users based on their requirements.
4. **Rapid elasticity:** Resources can be increased or decreased quickly according to demand.

Difference from traditional on-premise computing: Traditional computing requires organizations to purchase, maintain, and upgrade their own hardware. Cloud computing reduces the need for physical infrastructure and allows users to pay for the resources they use.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

The evolution of cloud computing took place through the following stages:

1. **Mainframe computing:** Large, centralized computers provided computing power to multiple users through terminals.
2. **Personal computers:** Computing became available to individual users, reducing dependence on centralized mainframes.
3. **Client-server computing:** Computers communicated over networks, sharing files, applications, and other resources.
4. **Virtualization:** A single physical server could run multiple virtual machines, improving hardware utilization and flexibility.
5. **Internet and web applications:** Faster internet connections enabled users to access applications and data remotely.
6. **Utility computing and cloud platforms:** Computing resources began to be provided as services, often with pay-as-you-use pricing.

Conclusion: Advances in hardware, virtualization, networking, and internet-based software enabled modern cloud platforms such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Server virtualization is a technology that allows one physical server to run multiple virtual machines, each with its own operating system and applications.

Role of server virtualization in cloud computing:

1. **Resource utilization:** It improves the use of physical server resources.
2. **Cost reduction:** It reduces the need for additional physical servers.
3. **Isolation:** Each virtual machine operates independently of others.
4. **Flexibility:** Virtual machines can be created, modified, or removed according to demand.
5. **Scalability:** It helps cloud providers allocate resources efficiently to different users.

Difference:

Virtualization	**Cloud Computing**
It is an enabling technology.	It is a service delivery model.
It creates virtual versions of computing resources.	It provides computing resources over a network.
It does not necessarily provide on-demand self-service or pay-as-you-use billing.	It commonly provides on-demand access and usage-based pricing.
Example: Running multiple virtual machines on one server.	Example: Renting virtual servers through AWS.

Conclusion: Virtualization helps cloud providers manage resources efficiently, while cloud computing delivers those resources as services to users.

4. Discuss the role of data centers in delivering cloud services.

A **data center** is a facility that contains servers, storage systems, networking equipment, and other infrastructure used to store, process, and manage data.

Role of data centers in cloud computing:

1. **Data storage:** They store large amounts of user and application data.
2. **Data processing:** Servers process requests and run cloud applications.
3. **Network connectivity:** Networking equipment connects servers to the internet and enables users to access cloud services.

4. **Security:** Data centers use access controls, monitoring, and other security measures to protect infrastructure and data.
5. **Availability and reliability:** Backup systems, redundant equipment, and disaster recovery mechanisms help maintain service availability.
6. **Scalability:** Data centers allow cloud providers to add computing and storage capacity as demand increases.

Examples: AWS, Microsoft Azure, and Google Cloud operate data center infrastructure to deliver cloud services worldwide.

Conclusion: Data centers form the physical foundation of cloud computing by providing the infrastructure required for storing data, processing requests, and delivering reliable services.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Cloud computing provides different service models according to users' requirements.

Service Model	Full Form	Description	Real-World Example
IaaS	Infrastructure as a Service	Provides virtual servers, storage, and networking. Users manage the operating system and applications.	Amazon EC2
PaaS	Platform as a Service	Provides a platform and tools for developing, testing, and deploying applications without managing the underlying infrastructure.	Google App Engine
SaaS	Software as a Service	Provides ready-to-use software through the internet, usually through a web browser or application.	Microsoft 365
XaaS	Anything as a Service	A broad term for delivering different IT resources and capabilities as services over a network.	Dropbox, which provides cloud storage as a service

Key differences:

- **IaaS:** Provides infrastructure.
- **PaaS:** Provides a development platform.
- **SaaS:** Provides ready-to-use software.
- **XaaS:** Covers a wide range of services delivered as needed.

Conclusion: These service models allow users to select cloud services according to their technical requirements, costs, and level of control.